

FRANKFORT, MI, USA

WMO: 720340

Lat: 44.626N

Lon: 86.201W

Elev: 193

StdP: 99.03

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.9	-20.1	0.6	-12.8	-17.7	0.8	-11.2	10.8	-5.3	9.6	-4.4	3.3	310	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.6	21.1	27.4	20.4	26.3	19.8	22.7	26.6	21.9	25.5	21.2	24.7	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.5	25.0	20.8	15.8	24.1	19.9	15.0	23.6	67.9	26.7	65.0	25.9	62.2	24.6	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.2	-19.1	31.1	3.5	1.1	-21.6	32.0	-23.7	32.6	-25.7	33.3	-28.2	34.1		
(5)			-19.3	24.1	3.6	0.8	-21.9	24.7	-24.0	25.2	-26.1	25.7	-28.7	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-3.9	-4.1	0.2	6.3	12.3	17.1	19.9	19.7	16.5	10.3	4.2	-0.8
	DBStd	9.74	5.00	5.06	6.02	4.84	4.78	3.57	3.24	2.80	4.09	4.60	4.48	4.32
	HDD10.0	1872	432	394	310	133	30	1	0	0	2	53	182	336
	HDD18.3	3898	690	627	562	362	198	64	22	18	83	253	424	594
	CDD10.0	1216	0	0	7	22	100	215	307	300	196	62	8	0
	CDD18.3	202	0	0	1	1	10	28	71	60	28	4	0	0
	CDH23.3	1332	0	0	2	2	79	204	504	359	166	15	0	0
	CDH26.7	187	0	0	0	0	11	28	86	39	24	0	0	0
Wind	WSAvg	2.9	3.7	3.4	2.9	3.0	2.6	2.2	2.1	2.1	2.4	3.1	3.6	3.6
Precipitation	PrecAvg	871	59	40	50	70	84	79	80	84	100	89	73	63
	PrecMax	1048	118	75	135	130	163	211	183	182	208	197	155	174
	PrecMin	577	15	13	10	20	24	19	17	23	21	24	21	14
	PrecStd	122	28	16	27	31	36	48	38	36	52	44	36	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	11.0	21.3	22.7	28.0	29.0	31.0	29.2	29.0	25.2	18.8	13.7
		MCWB	7.1	7.3	15.5	13.6	18.9	20.7	22.3	21.4	20.6	19.3	13.5	12.0
	2%	DB	6.4	7.4	16.2	19.9	25.9	27.4	28.8	27.7	27.2	22.9	15.7	8.0
		MCWB	4.7	5.1	12.1	11.8	18.5	19.4	21.4	20.7	20.7	17.7	11.4	6.6
	5%	DB	4.3	4.4	12.0	17.3	23.0	26.0	27.4	26.6	25.3	20.3	12.8	6.8
		MCWB	2.8	2.8	8.0	10.3	16.2	18.8	20.7	20.3	19.4	16.0	9.8	5.2
	10%	DB	2.8	2.6	8.2	14.5	20.8	24.0	26.2	25.3	23.5	17.7	11.1	5.0
		MCWB	1.7	1.0	4.9	8.7	14.8	18.0	20.0	19.2	18.8	14.1	8.3	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	8.2	15.9	16.2	21.1	23.1	24.2	23.5	23.2	20.7	14.2	12.0
		MCDB	8.6	9.7	20.8	21.1	26.3	28.0	28.8	27.0	26.7	24.2	17.0	13.5
	2%	WB	5.0	5.1	12.3	13.2	19.4	21.0	22.8	22.4	21.9	19.2	12.1	7.4
		MCDB	6.2	7.2	14.9	18.3	24.5	25.4	27.1	25.8	25.4	22.3	14.7	7.9
	5%	WB	3.0	3.0	7.9	11.1	17.6	19.9	21.9	21.7	20.9	16.7	10.2	5.3
		MCDB	4.0	4.4	10.6	16.4	22.0	24.3	25.7	24.9	24.0	19.2	12.9	6.6
	10%	WB	1.8	1.1	5.2	9.2	15.7	18.9	21.1	20.8	19.7	14.5	8.5	3.6
		MCDB	2.9	2.3	8.0	13.8	19.7	22.9	24.8	23.9	22.9	17.8	10.7	4.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.2	8.4	9.3	10.7	10.4	10.0	9.6	9.9	8.1	6.2	4.8
		MCDBR	7.3	8.1	11.4	13.3	13.0	12.0	10.8	10.6	10.8	10.7	9.3	7.3
		MCWBR	6.6	6.9	7.9	7.9	7.4	6.3	5.5	5.8	5.9	7.2	6.7	6.9
	5% WB	MCDBR	6.8	7.7	10.9	11.8	11.6	9.9	8.8	8.4	8.7	9.2	8.2	7.4
		MCWBR	6.3	6.8	8.3	8.1	7.3	6.0	5.1	5.4	5.6	7.4	6.9	7.1
Clear-Sky Solar Irradiance	taub	0.289	0.300	0.324	0.361	0.389	0.403	0.401	0.394	0.378	0.344	0.313	0.285	
	taud	2.459	2.403	2.414	2.353	2.347	2.345	2.365	2.382	2.414	2.484	2.508	2.497	
	Ebn at Noon	838	891	913	901	882	867	864	858	844	829	802	808	
	Edh at Noon	70	89	101	117	121	122	119	113	101	82	67	62	
All-Sky Solar Radiation	RadAvg	1.10	2.02	3.51	4.63	5.68	6.18	6.27	5.31	4.11	2.45	1.34	0.86	
	RadStd	0.12	0.22	0.38	0.43	0.48	0.43	0.27	0.34	0.36	0.28	0.19	0.10	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page